§3.8 Derivatives of Logarithmic Functions

Homework：3,9,17,21,28,35,39,41,43,48,50

3.8 節的一些公式.

(I) $\frac{d}{dx}\left(\log_a x\right)=\frac{1}{x\ln a}$.

$\frac{d}{dx}\ln x=\frac{1}{x}$.

$\frac{d}{dx}\ln|x|=\frac{1}{x}$.

(II) $\left(x^r\right)'=rx^{r-1}$, $r\in R$.

(III) $e=\lim\limits_{x\to 0}\left(1+x\right)^{\frac{1}{x}}=\lim\limits_{x\to\infty}\left(1+\frac{1}{n}\right)^n$.

$e^x=\lim\limits_{x\to\infty}\left(1+\frac{x}{n}\right)^n$, $x>0$.

Why？

(I) $y=\log_a x$

$\Rightarrow a^y=x\Rightarrow a^y\left(\ln a\right)y'=1\Rightarrow y'=\frac{1}{\ln a}\frac{1}{a^y}=\frac{1}{\ln a}\frac{1}{x}$.

if $x<0$, $\ln|x|=\ln(-x)\Rightarrow\frac{d}{dx}\ln|x|=\frac{d}{dx}\ln(-x)=\frac{-1}{-x}=\frac{1}{x}$.

if $x>0$, $\ln|x|=\ln x\Rightarrow\frac{d}{dx}\ln|x|=\frac{d}{dx}\ln x=\frac{1}{x}$.

(II) $y=x^r\Rightarrow\ln|y|=\ln|x|^r=r\ln|x|$

$\Rightarrow\frac{y'}{y}=\frac{r}{x}\Rightarrow y'=r\frac{y}{x}=rx^{r-1}$.

(III) Let $f(x)=\ln x \Rightarrow f'(x)=\dfrac{1}{x} \Rightarrow f'(1)=1.$

$$1=f'(1)=\lim_{x\to 0}\frac{f(1+x)-f(1)}{x}=\lim_{x\to 0}\frac{\ln(1+x)}{x}=\lim_{x\to 0}\frac{1}{x}\ln(1+x)$$

$$=\lim_{x\to 0}\ln(1+x)^{\frac{1}{x}}=\ln\lim_{x\to 0}(1+x)^{\frac{1}{x}}$$

$$\Rightarrow \lim_{x\to 0}(1+x)^{\frac{1}{x}}=e.$$

(最後一個等式用到連續函數和極限可交換的性質)

Example 1： $y=\log\left(x^3+1\right)$

$$\Rightarrow y'=\frac{3x^2}{(\ln 10)\left(x^3+1\right)}.$$

Example 2： $y=\dfrac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}$

$$\Rightarrow \ln y=\frac{3}{4}\ln x+\frac{1}{2}\ln\left(x^2+1\right)-5\ln(3x+2)$$

$$\Rightarrow \frac{y'}{y}=\frac{3}{4x}+\frac{x}{x^2+1}-\frac{15}{3x+2}$$

$$\Rightarrow y'=\left(\frac{3}{4x}+\frac{x}{x^2+1}-\frac{15}{3x+2}\right)y.$$

Example 3： $y=x^x$, *Find* $\dfrac{dy}{dx}$. (若 $y=x^2$, 則 $y'=2x$. 若 $y=2^x$, 則 $y'=2^x\ln 2$)

Solution： $\ln y=x\ln x$

$$\Rightarrow \frac{y'}{y}=\ln x+1 \Rightarrow y'=x^x(1+\ln x).$$

Example 4： $y=x^{\cos x}$.

$$\Rightarrow \ln y = \cos x \ln x$$

$$\Rightarrow \frac{y'}{y} = -\sin x \ln x + \frac{\cos x}{x}$$

$$y' = \left(-\sin x \ln x + \frac{\cos x}{x} \right) x^{\cos x}.$$

Example 5： $\dfrac{d^9}{dx^9}\left(x^8 \ln x\right)$

$$= \frac{d^8}{dx^8}\left(8x^7 \ln x + x^7\right) = D^8\left(8x^7 \ln x\right)$$

$$= D^7\left(8\times 7x^6 \ln x + 8x^6\right) = D^7\left(8\times 7x^6 \ln x\right)$$

$$= D\left(8!\, x^0 \ln x\right) = \frac{8!}{x}.$$